UNIT I

Introduction to Composite Materials & FRP Processing

Definition | classification | applications | lay-up, bag moulding, filament winding, pultrusion, moulding

I. Introduction & Definition

A **composite material** is a material made by combining two or more distinct constituents (different in form or chemical composition, and insoluble in each other) on a **macroscopic scale** so that the combination has properties that are better than those of the individual constituents acting alone.

Examples: wood (cellulose fibres in lignin), bone, reinforced concrete (steel in concrete), glass-fibre/epoxy, carbon/epoxy, Al-SiC.

Why study composites ?

> Metals are isotropic and heavy; modern aircraft, space and sports applications need high **strength-to-weight** and **stiffness-to-weight**.

> Composites let the designer **tailor** the material: choose fibre, matrix, volume fraction and fibre direction to suit the load.

> Properties are *not* fixed: they depend on the design ($E_c \neq E_f, E_m$) - material and structure are made together.

I.1 Constituents of a composite

- **Reinforcement (dispersed phase)** - fibres, particles or flakes. Carries the major load, gives strength and stiffness.
- **Matrix (continuous phase)** - polymer, metal or ceramic. Binds the fibres, transfers load between them, protects them from environment and keeps them in position.
- **Interface** - the bond region between fibre and matrix; governs load transfer, toughness and moisture resistance.

Composite = Reinforcement + Matrix + Interface

I.2 Functions of the matrix

- Holds the fibres together and gives the component its shape.
- Transfers the applied load to the fibres through shear at the interface.

- Separates the fibres, so a crack in one fibre does not run into the next (crack arrest).
- Protects fibres from abrasion, moisture, heat and chemicals.
- Controls inter-laminar shear strength, transverse strength and the maximum service temperature.

1.3 Functions of the reinforcement

- Provides the strength and stiffness: in a good composite fibres carry 70-90 % of the load.
- Controls thermal expansion, thermal and electrical conductivity.
- Gives wear and abrasion resistance (particles, whiskers).

II. Characteristics of Composite Materials

Advantages	Limitations
High specific strength (σ/ρ) and specific modulus (E/ρ)	High raw-material and fabrication cost
Light weight: 20-50 % mass saving in aircraft	Properties vary with batch / process (scatter)
Excellent fatigue and corrosion resistance	Brittle, low impact and transverse strength
Tailorable (anisotropic) properties, directional design	Difficult to repair and to recycle
Low thermal expansion, dimensional stability	Moisture absorption and UV / heat degradation
Part integration, complex shapes in one moulding	Complex analysis; delamination, hard to inspect

2.1 Anisotropy and heterogeneity

Composites are **heterogeneous** (properties change from point to point - fibre or matrix) and usually **anisotropic** (properties depend on direction). A unidirectional lamina is very strong along the fibres (direction 1) but weak across them (direction 2).

E_1 (along fibre) $\gg E_2$ (transverse)

e.g. carbon/epoxy: $E_1 \approx 140$ GPa, $E_2 \approx 10$ GPa

Specific properties (comparison)

> Steel: $\rho = 7.8$ g/cc, $E = 210$ GPa → $E/\rho \approx 27$
> Aluminium: $\rho = 2.7$ g/cc, $E = 70$ GPa → $E/\rho \approx 26$
> Carbon/epoxy: $\rho = 1.6$ g/cc, $E_1 = 140$ GPa → $E/\rho \approx 88$ (3 times better)

III. Classification of Composite Materials

Composites are classified (a) by **matrix**: PMC, MMC, CMC, carbon-carbon; and (b) by **reinforcement shape**: fibrous, laminated and particulate.

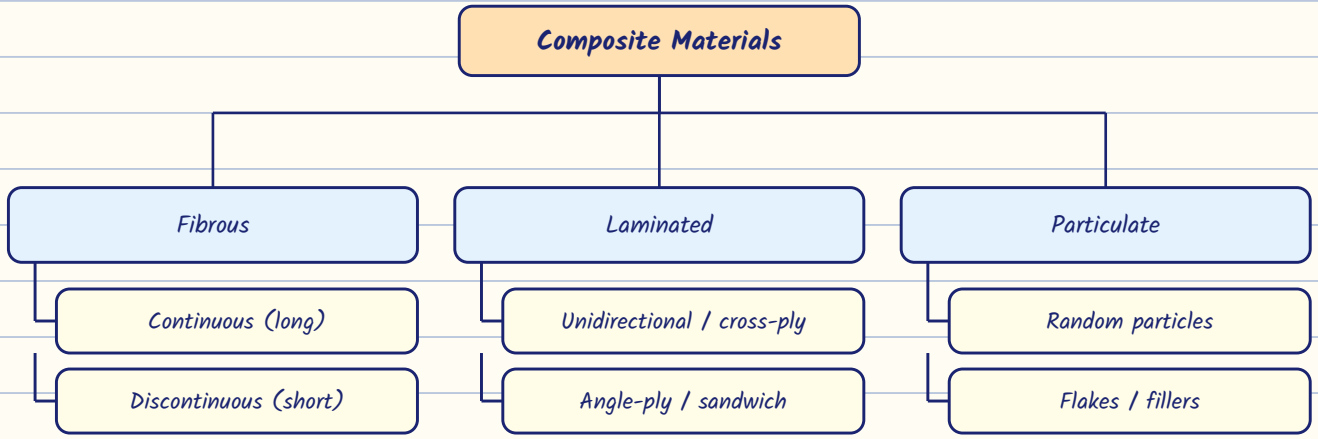


Fig. 1 Classification of composites based on reinforcement

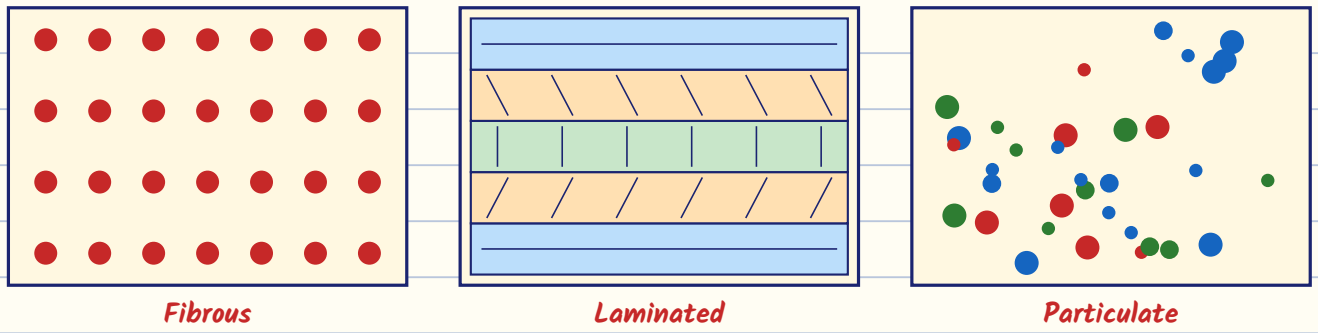


Fig. 2 Sketch of fibrous, laminated and particulate composites

3.1 Fibrous composites

Fibres (length ≫ diameter) in a matrix. **Continuous-fibre** composites (unidirectional, woven, braided) give the highest strength; **discontinuous (short, chopped)** fibres can be aligned or random - lower strength but easy to mould. Strength depends on fibre length, orientation and volume fraction V_f.

3.2 Laminated composites

Made of **layers (plies / laminae)** bonded together, each lamina having its own fibre orientation (0°, ±45°, 90° ...). The stacking sequence is chosen to get the required stiffness in several directions. Also includes **sandwich panels** (thin stiff face sheets + light core of honeycomb or foam) and clad metals, bimetals, plywood.

3.3 Particulate composites

Particles (spherical, flake, irregular) of one material in a matrix of another. Properties are roughly **isotropic**. Used to improve wear, hardness, heat resistance and stiffness: e.g. concrete (sand/gravel in cement), cermets (WC-Co cutting tools), Al-SiC pistons, rubber with carbon black, filler-loaded plastics ($CaCO_3$, talc).

Type	Reinforcement	Main feature	Example
Fibrous	Long / short fibres	Strong in fibre direction	GFRP, CFRP
Laminated	Stacked plies / layers	Tailored in-plane stiffness	Plywood, CFRP laminate
Particulate	Particles, flakes	Isotropic, wear / hardness	Concrete, Al-SiC

IV. Reinforcements and Matrices

4.1 Common fibres

Fibre	Density (g/cc)	Tensile strength (GPa)	Modulus (GPa)	Remarks
E-glass	2.55	2.4-3.4	72	Cheap, most used, good insulator
S-glass	2.50	4.5	86	Higher strength, aerospace
Carbon (HS / HM)	1.75-1.95	3.5-5.5 / 2.5	230 / 400-600	Stiff, light; costly, galvanic with Al
Aramid (Kevlar-49)	1.45	3.6	130	Tough, impact / ballistic; poor in compression
Boron	2.6	3.5	400	Very stiff, big diameter, very costly
SiC / Alumina	2.5-3.9	2-3	300-400	High-temp., MMC / CMC

4.2 Matrix materials

Property	Thermoset (epoxy, polyester, vinylester, phenolic)	Thermoplastic (PEEK, nylon, PP, PPS)
Curing	Irreversible chemical cross-linking; cannot be re-melted	No chemical change; softens on heating, re-formable
Viscosity	Low: easy fibre wet-out	High: difficult impregnation
Properties	Good creep, high temp. stability	Tough, good impact, recyclable
Processing	Long cure cycle (hours)	Fast cycle (minutes), high temperature

Remember: Polyester is cheapest (general GFRP); **epoxy** is the standard aerospace matrix (better adhesion, low shrinkage, 120-180 °C service).

V. Applications of Composites

5.1 Automobile

- Body panels, bumpers, bonnets, doors (SMC / GFRP) - light weight, no rusting, easy complex shapes.
- Drive shafts (CFRP/GFRP), leaf springs, wheels, connecting rods, brake pads, clutch plates.
- F1 monocoque and crash structures (carbon/epoxy); electric-vehicle battery casings. 10 % weight saving → 6-8 % fuel saving.

5.2 Aircraft

- Fuselage, wings, tail, rudder, elevator, floor beams and fairings: composites ≈ 50 % of airframe weight in modern airliners (B787, A350).
- Helicopter rotor blades, propellers, landing-gear doors, radomes, interior panels.
- Military aircraft: carbon/epoxy skins, radar-absorbing (stealth) structures.

5.3 Missiles

- Rocket motor cases (filament wound Kevlar / carbon-epoxy), nose cones, radomes.
- Ablative heat shields (carbon-phenolic), nozzles and launch tubes - high strength/weight, heat resistance, transparency to radar.

5.4 Space hardware

- Satellite structures, antenna reflectors, solar-panel substrates, truss booms - dimensional stability (near-zero CTE of carbon/epoxy).
- Space-shuttle tiles & nose cap (carbon-carbon), payload bay doors, launch-vehicle fairings, pressure vessels.

5.5 Electrical and electronics

- Printed circuit boards (glass/epoxy FR-4), insulators, switch-gear panels, transformer housings, cable trays.
- Electromagnetic shielding (metal-coated fibres), heat sinks (Al-SiC, Cu-C), antenna radomes, wind-turbine blades.

5.6 Marine

- Boat hulls, decks, masts, yachts, canoes, offshore platforms and pipes - corrosion resistance, no rotting, low maintenance.
- Mine-sweepers (non-magnetic), submarine hulls and sonar domes, propellers, storage tanks.

5.7 Recreational and sports equipment

- Tennis and badminton rackets, golf shafts, hockey sticks, fishing rods, archery bows, bicycle frames.
- Skis, snowboards, vaulting poles, cricket bats, helmets, prosthetic limbs: light, stiff and damping.

5.8 Future potential of composites

- **Nano-composites** (CNT, graphene) - very high strength, conductivity; **bio-composites** (natural fibres: jute, flax, hemp) - green and recyclable.
- **Smart / multifunctional** composites with embedded sensors and actuators, self-healing resins; **thermoplastic** composites for fast automated production and recycling.
- Growth in hydrogen tanks, wind & renewable energy, civil infrastructure (bridge decks, rebars), 3D-printed and automated fibre placement for low cost.

VI. Fibre Reinforced Plastic (FRP) Processing

Processing converts fibres + resin into a finished part. The basic steps in every method are: **(1) fibre placement → (2) resin impregnation (wet-out) → (3) shaping on a mould → (4) curing / consolidation → (5) demoulding and finishing.**



Fig. 3 General sequence of FRP processing

6.1 Lay-up and curing

- **Lay-up**: reinforcement (mat, woven roving, fabric, prepreg) is placed ply by ply on the mould in the specified orientation; resin is applied and worked in to remove air.
- **Curing**: the resin changes from liquid to solid by cross-linking, an **exothermic** reaction started by a catalyst / hardener, heat or UV. Cure may be at room temperature (hours/days) or at elevated temperature (80-180 °C) in an oven or autoclave.

- Cure variables: temperature, time, pressure. **Post-cure** improves cross-linking and glass-transition temperature T_g.

6.2 Fabricating process: open mould and closed mould

Open-mould process	Closed-mould process
One mould surface; only one side finished (smooth)	Two mould halves (or mould + bag); both sides finished
Low tooling cost, large parts (boats, tanks)	High tooling cost, high volume small / medium parts
Hand lay-up, spray-up, (wet lay-up, bag-moulding)	Compression, injection, RTM, pultrusion, resin infusion
Slow, labour intensive, emission of styrene	Fast cycle, good dimensional control, low emission
Lower fibre volume fraction (30-40 %)	Higher fibre volume fraction (up to 60-70 %)

6.3 Hand lay-up technique

The oldest, simplest and cheapest method; used for boat hulls, tanks, roofing sheets, bath tubs.

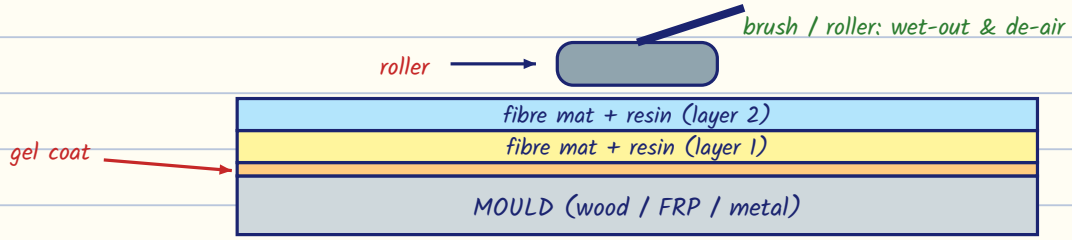


Fig. 4 Hand lay-up

- **Step 1** - clean the mould and apply **release agent** (wax / PVA).
- **Step 2** - spray or brush the **gel coat** (pigmented resin, 0.3-0.5 mm) for a smooth, weather-proof surface; let it gel.
- **Step 3** - place the first glass **mat / woven roving**, pour resin (+ catalyst, accelerator) and **wet it out** with brush or roller.
- **Step 4** - consolidate with a serrated **roller** to squeeze out air bubbles; add layers until thickness is obtained.
- **Step 5** - cure at room temperature (or low heat), demould, trim, finish.

Advantages of hand lay-up	Disadvantages
Low cost tooling, simple equipment	Quality depends on operator skill

Advantages of hand lay-up	Disadvantages
Large and complex shapes, any size	Only one smooth face; slow, labour intensive
Wide choice of fibres and resins	Low V_p porosity, resin-rich areas, health hazard

6.4 Spray-up

Chopped fibre (25-50 mm) and catalysed resin are sprayed together by a **chopper gun** onto the mould, then rolled out. Faster than hand lay-up, fibre is random (short), lower strength - used for bath tubs, panels, containers.

VII. Structural Laminate Bag Moulding

To get higher fibre content and fewer voids than hand lay-up, the laid-up laminate is covered with a flexible **bag** and **pressure** is applied (by vacuum, air or steam) during cure. Used for aircraft skins, radomes, sports goods.

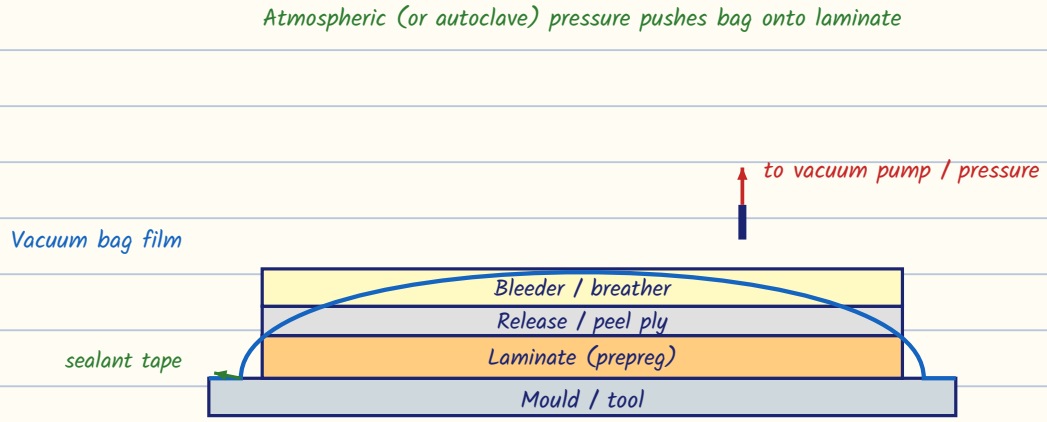


Fig. 5 Vacuum bag moulding (cross-section)

7.1 Types of bag moulding

Process	How pressure is applied	Typical pressure
Vacuum bag	Air evacuated from under bag; atmosphere presses it	≈ 0.1 MPa (1 atm)
Pressure bag	Compressed air / steam inside a bag over the lay-up	0.2-0.4 MPa
Autoclave	Vacuum bag + heated pressure vessel (prepreg)	0.5-0.7 MPa, 120-180 °C

7.2 Production procedure for bag moulding

- 1. Prepare mould surface, apply release agent / gel coat.
- 2. Lay up prepreg or wet plies in the correct stacking sequence and orientation.

- 3. Cover with **peel ply** (release film) → **bleeder / breather** cloth (absorbs excess resin, allows air path).
- 4. Place the **vacuum bag** film, seal edges with sealant tape, connect the vacuum port.
- 5. Apply vacuum (and heat / pressure); **debulk** and cure as per cycle (ramp - dwell - cool).
- 6. Release pressure, remove bag, bleeder and peel ply, demould, trim and inspect.

Exam tip: Learn the bag stack (mould → release → laminate → peel ply → bleeder → breather → bag) and the 3 types of bag moulding with sketches.

VIII. Filament Winding

A continuous process in which **resin-impregnated fibre (roving / tow)** is wound under tension over a rotating **mandrel** while a carriage traverses along the axis. Used for hollow, axisymmetric parts: pipes, pressure vessels, gas cylinders, rocket motor cases, drive shafts.

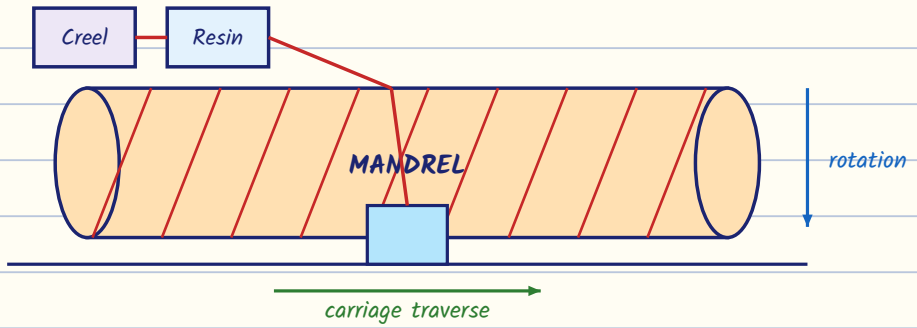


Fig. 6 Filament winding set-up

8.1 Winding patterns

- **Hoop (circumferential) winding** - angle ≈ 90° to the axis; carries hoop stress.
- **Helical winding** - angle 15°-75°; carries both hoop and axial stress.
- **Polar winding** - fibre passes tangent to the polar openings; for spherical / dome vessels.

8.2 Steps, advantages and limitations

- Steps: fibre creel → resin bath (wet winding) or prepreg tow → delivery eye on carriage → winding on mandrel → cure (oven / rotating) → **mandrel extraction** → finishing.
- **Advantages**: high V_f (60-70 %), precise fibre orientation, automation, low cost of fibre (roving), high strength-to-weight.
- **Limitations**: only convex / axisymmetric shapes, mandrel cost, poor external surface, fibre cannot be laid along the axis easily.

Netting analysis (helical angle)

> For a closed pressure vessel: hoop stress $\sigma_\theta = pr / t$, axial $\sigma_x = pr / 2t$.

> Fibre-only theory gives the helical angle $\alpha = \tan^{-1}\sqrt{2} \approx$ **54.7°** from the axis for equal load sharing.

IX. Pultrusion and Pulforming

Pultrusion is a continuous process to make constant-cross-section profiles (rods, channels, I-beams, tubes). Fibres are **pulled** (not pushed) through a resin bath and then through a **heated die** where the resin cures; a puller grips the cured profile and a saw cuts to length.

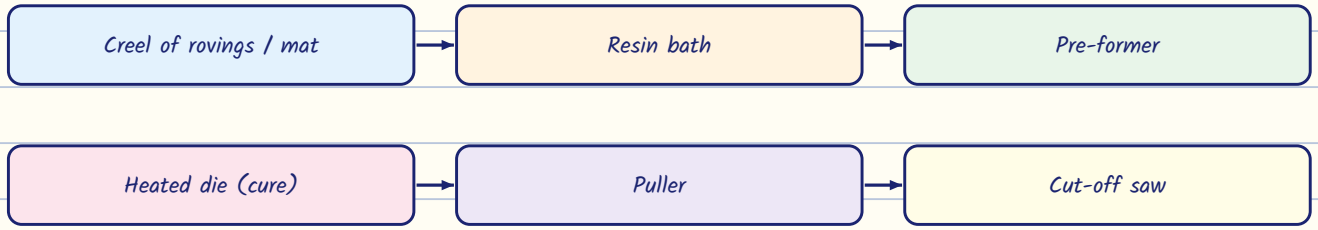


Fig. 7 Pultrusion line

- **Advantages**: continuous, fully automatic, low labour, V_f up to 70 %, excellent axial strength, low scrap.
- **Limitations**: only constant cross-section; heated die cost; low transverse strength unless mat is used.
- Uses: ladders, window frames, cable trays, bridge decks, fishing rods, electrical insulators, antenna poles.

9.1 Pulforming

A variation of pultrusion for **non-uniform / curved profiles**: after impregnation, the fibre profile is pulled through a **moving, closed mould** (or a die and a forming station) that shapes it into a curved or varying section. Used for leaf springs, curved handles, automotive parts.

X. Thermoforming, Injection and Blow Moulding

These are the main processes for **thermoplastic** matrix composites and filled plastics.

10.1 Thermoforming

- A heated **thermoplastic composite sheet** (organosheet) is softened, then shaped over / into a mould by **vacuum, pressure or matched dies** and cooled to set.
- Fast (30-60 s), recyclable; used for automotive panels, trays, aircraft interior, helmets.

10.2 Injection moulding

- Pellets of **short-fibre-filled thermoplastic** are melted in a heated barrel and **injected under high pressure** by a reciprocating screw into a closed, cooled mould; part is ejected after cooling.
- Cycle time seconds; very good for mass production of complex small parts (gears, housings, connectors).
- Limitations: fibre breakage → short fibre ($L < 1$ mm), low strength, anisotropy due to flow, expensive moulds.

10.3 Blow moulding

- A hot hollow tube of thermoplastic (**parison**) is clamped in a mould and inflated by compressed air against the mould walls, then cooled.
- Used for bottles, containers, fuel tanks and ducts; reinforced grades use short glass fibres or nano-fillers.

Process	Matrix	Product	Production
Hand lay-up	Thermoset	Boat, tank	Low
Bag moulding / autoclave	Thermoset prepreg	Aircraft skin	Low-medium
Filament winding	Thermoset	Pipes, vessels	Medium
Pultrusion	Thermoset	Profiles, rods	High (continuous)
Thermoforming	Thermoplastic	Panels, trays	High
Injection / blow	Thermoplastic	Small parts, bottles	Very high

XI. Quick Revision - Unit I

Important points

> Composite = matrix + reinforcement; **classification**: fibrous, laminated, particulate.
> Advantages: high specific strength / stiffness, tailoring, fatigue and corrosion resistance. Limitations: cost, brittleness, repair.
> Open mould = hand lay-up, spray-up; closed mould = compression, injection, RTM, pultrusion.
> Bag moulding: vacuum / pressure / autoclave; stack = release, peel ply, bleeder, breather, bag.
> Filament winding: hoop / helical / polar → pressure vessels; pultrusion: continuous profiles; pulforming: curved profiles.

Likely exam questions

> Define composite materials. Classify and compare fibrous, laminated and particulate composites.
> Explain hand lay-up technique with a neat sketch, merits and demerits.
> Describe structural laminate bag moulding and its production procedure.
> With neat sketches explain filament winding and pultrusion. Write short note on pulforming.
> Write applications of composites in automobile, aircraft, marine and sports.

Short Answer Questions (2 marks)

Q1. Define a composite material.

Ans: A material made of two or more constituents differing in form / composition, combined on a macroscopic scale, with properties better than the individual ones.

Q2. Name the two phases of a composite.

Ans: Continuous **matrix** and dispersed **reinforcement**.

Q3. What is specific strength?

Ans: Strength divided by density (σ/ρ); specific modulus = E/ρ.

Q4. Differentiate thermoset and thermoplastic matrix.

Ans: Thermoset cross-links irreversibly on curing (epoxy, polyester); thermoplastic softens on heating and can be reshaped (PEEK, nylon).

Q5. What is a prepreg?

Ans: Fibres (UD or fabric) pre-impregnated with partially cured resin; stored cold and cured in an autoclave.

Q6. Purpose of gel coat?

Ans: Smooth, pigmented, weather and water resistant outer surface in hand lay-up.

Q7. Why is a bleeder cloth used in bag moulding?

Ans: Absorbs excess resin so that fibre volume fraction increases and voids are reduced.

Q8. Which process makes pipes and pressure vessels?

Ans: Filament winding.

Q9. What is pultrusion?

Ans: Continuous pulling of resin-wet fibres through a heated die to form constant cross-section profiles.

Q10. Difference between pultrusion and pulforming?

Ans: Pulforming makes curved / varying cross-sections using a moving mould; pultrusion makes straight constant sections.

Q11. Give two limitations of composites.

Ans: High cost, brittleness / poor impact resistance, difficult repair and recycling, moisture absorption.

Glossary of Key Terms

Term	Meaning
Prepreg	Pre-impregnated fibre sheet with B-stage resin
Roving	Bundle of continuous untwisted strands wound on a spool
Mat	Randomly oriented chopped or continuous strand sheet
Debulking	Compacting plies under vacuum to remove air before cure
Gel time	Time for liquid resin to turn to a gel after catalyst is added
Autoclave	Heated pressure vessel for curing aerospace laminates
SMC / BMC	Sheet / bulk moulding compound for compression moulding
RTM	Resin transfer moulding: resin injected into a closed mould containing dry fibres

Answer Outlines for 10-mark Questions

Hand lay-up technique

- Sketch: mould, gel coat, plies, roller.
- Steps: release agent → gel coat → mat + resin → rolling → cure → demould.
- Advantages: low tooling cost, large parts; disadvantages: low V_f, operator dependent.
- Applications: boats, tanks, bath-tubs.

Filament winding

- Sketch of mandrel, carriage, resin bath, creel.
- Patterns: hoop, helical, polar; winding angle selection (54.7°).
- Cure, mandrel extraction; advantages and limitations; uses.

Bag moulding

- Lay-up stack and sketch of vacuum bag.
- Vacuum, pressure and autoclave variants with pressures.
- Production procedure steps; merits: higher V_f, lower voids.